

RAJALAKSHMI ENGINEERING COLLEGE

An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(CYBER SECURITY)

CR23532- Cyber and Digital Forensics

LAB MANUAL

2023 REGULATION

FIFTH SEMESTER

III YEAR

EXPERIMENT 1

STUDY OF COMPUTER FORENSICS AND DIFFERENT TOOLS USED FOR FORENSIC INVESTIGATION

AIM:

To study computer forensics, understand the digital forensic investigation process, and examine the different tools used for forensic investigation.

SOFTWARE REQUIREMENTS:

- Operating System: Windows / Linux
- Digital forensic tools: Autopsy, FTK Imager, The Sleuth Kit, and other forensic utilities.

THEORY:

Computer forensics, also known as digital forensics, is the process of identifying, collecting, preserving, examining, and analyzing digital evidence from computers and other digital devices. It is used to investigate cybercrimes, security incidents, unauthorized access, and other activities involving digital systems.

Digital forensic investigators examine data from storage devices, operating systems, application logs, browser histories, deleted files, and other sources to identify evidence relevant to an investigation.

The primary objective of digital forensics is to recover and analyze digital evidence while maintaining its integrity and ensuring that the findings can be presented accurately.

DIGITAL FORENSICS PROCESS:

The digital forensic investigation process consists of the following stages:

1. Identification

Identifying potential sources of digital evidence, such as computers, hard disks, USB drives, mobile devices, and log files.

2. Collection

Collecting relevant digital evidence using appropriate forensic procedures and tools.

3. Preservation

Protecting the collected evidence from unauthorized modification or damage. Investigators may create forensic images and calculate hash values to help verify evidence integrity.

4. Examination

Examining the collected evidence to locate relevant files, metadata, deleted data, and other artifacts.

5. Analysis

Analyzing the extracted information to establish relationships between events, identify suspicious activities, and develop findings relevant to the investigation.

6. Presentation

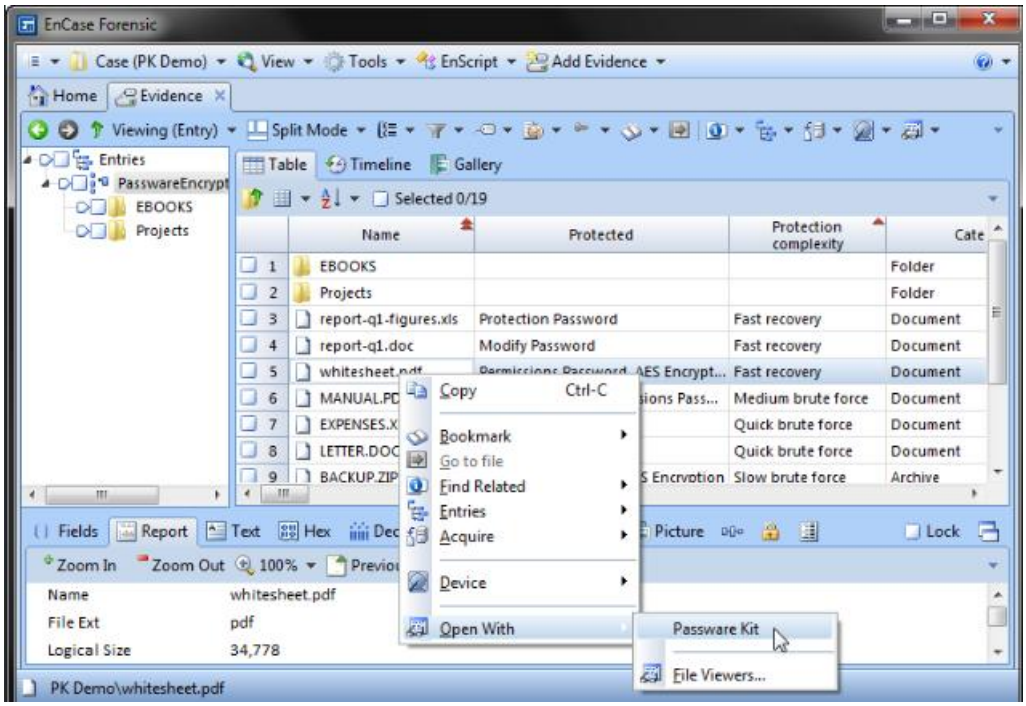
Documenting the investigation methods, observations, and conclusions in a report suitable for technical or legal review.

DIFFERENT TOOLS USED IN COMPUTER FORENSICS

The lab manual discusses commercial forensic suites and free or open-source tools used in digital investigations.

A. EnCase Forensic

EnCase Forensic is a commercial digital forensic investigation tool used to acquire, examine, and analyze digital evidence. It supports forensic investigations involving computers and storage devices.



B. Forensic Toolkit (FTK)

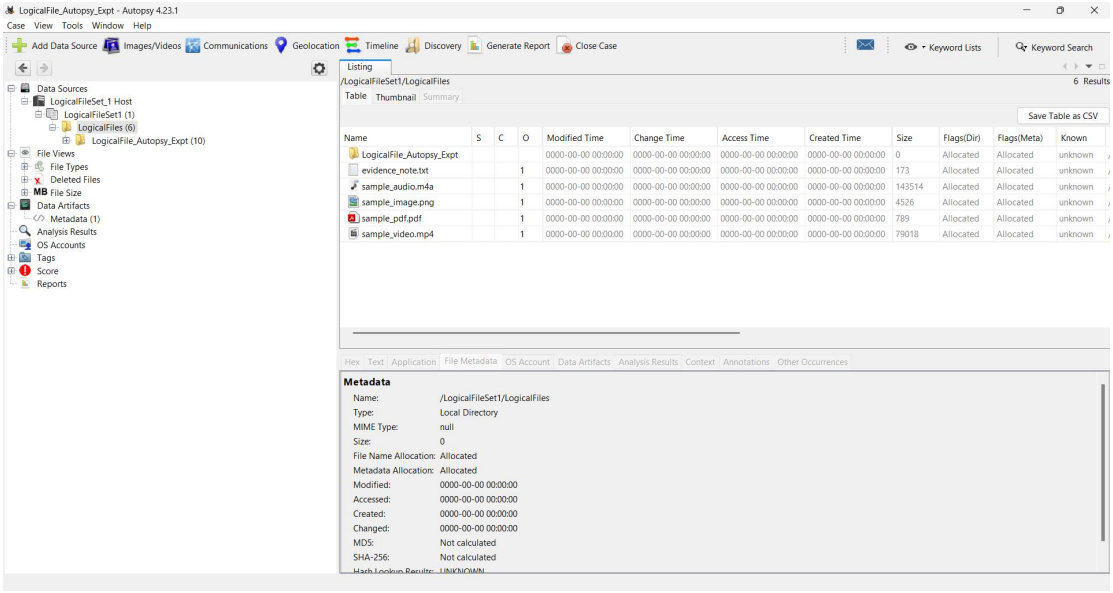
FTK is a digital forensic investigation suite used to analyze digital evidence. It supports searching, file analysis, and examination of information collected during investigations.

C. The Sleuth Kit (TSK)

The Sleuth Kit is an open-source collection of command-line tools used to analyze disk images and file systems. It helps investigators examine file-system structures and recover information relevant to an investigation.

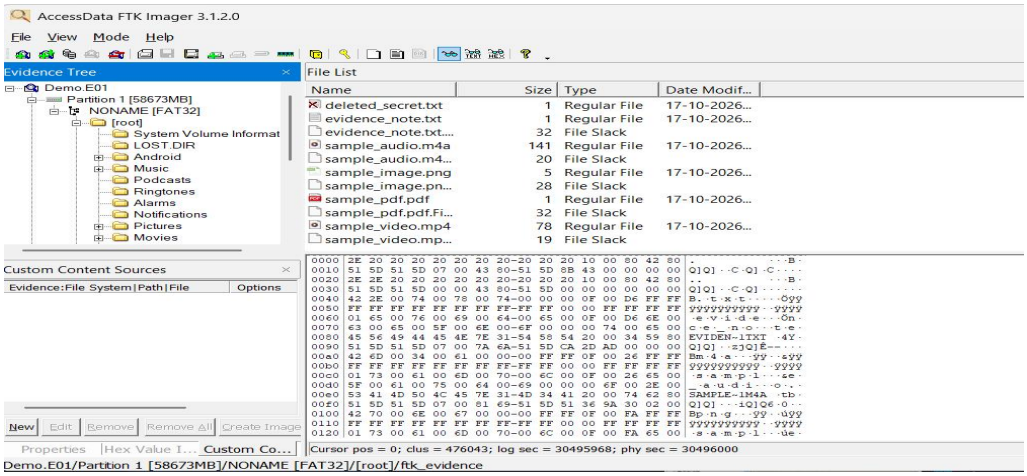
D. Autopsy

Autopsy is a digital forensic platform that provides a graphical interface for The Sleuth Kit and other forensic tools. It supports the examination of disk images, file recovery, keyword searching, and analysis of digital artifacts.



E. FTK Imager

FTK Imager is a forensic imaging utility used to acquire and examine forensic images of storage devices. It can also help investigators inspect files and folders contained in supported evidence images.



APPLICATIONS OF COMPUTER FORENSICS:

Digital forensics is used in several areas:

1. **Cybercrime investigation:** Examining digital evidence associated with unauthorized access, malware, and other cyber incidents.
2. **Data recovery:** Recovering deleted or inaccessible files where possible.
3. **Incident response:** Investigating security incidents and determining their possible causes.
4. **Employee misconduct investigation:** Examining relevant digital artifacts during authorized workplace investigations.
5. **Legal investigations:** Preserving and analyzing digital evidence for potential use in legal proceedings.
6. **Network investigation:** Examining network logs and related artifacts to understand suspicious communications and activities.

GENERAL PROCEDURE:

1. Study the fundamental concepts of computer and digital forensics.
2. Identify the major stages of the digital forensic investigation process.
3. Study the different commercial and open-source forensic tools.
4. Understand the purpose and functions of tools such as FTK Imager, Autopsy, and The Sleuth Kit.
5. Examine the applications of digital forensics in cybercrime investigation and data recovery.
6. Document the observations and findings.

OBSERVATION:

The study provided an understanding of computer forensics and the importance of digital evidence in cybercrime investigations.

Different forensic tools were examined based on their purposes and capabilities. The investigation process was understood to involve identifying, collecting, preserving, examining, analyzing, and presenting digital evidence.

RESULT:

Thus, the study of computer forensics and the different tools used for forensic investigation was completed successfully.

EXPERIMENT 2

LIVE FORENSICS CASE INVESTIGATION USING AUTOPSY

AIM:

To perform a digital forensic investigation using Autopsy by creating a case, adding a data source, analyzing digital evidence, and examining the forensic artifacts generated by the tool.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- Software: Autopsy Digital Forensic Tool
- Evidence source: A suitable forensic image or disk image
- Hardware: Computer with sufficient storage and memory

THEORY:

Autopsy is a digital forensic investigation platform that provides a graphical interface for examining digital evidence. It uses The Sleuth Kit and other tools to help investigators analyze files, disk images, and digital artifacts.

Autopsy supports activities such as file-system analysis, keyword searching, recovery of deleted files, and examination of information related to user activities.

In a forensic investigation, a case is created to organize the investigation details and associated evidence. A data source, such as a disk image, is then added to the case. Autopsy processes the source using selected ingest modules and displays the extracted information for examination.

The results can help investigators identify relevant files, user accounts, browser activity, metadata, and other artifacts associated with the evidence source.

PROCEDURE:

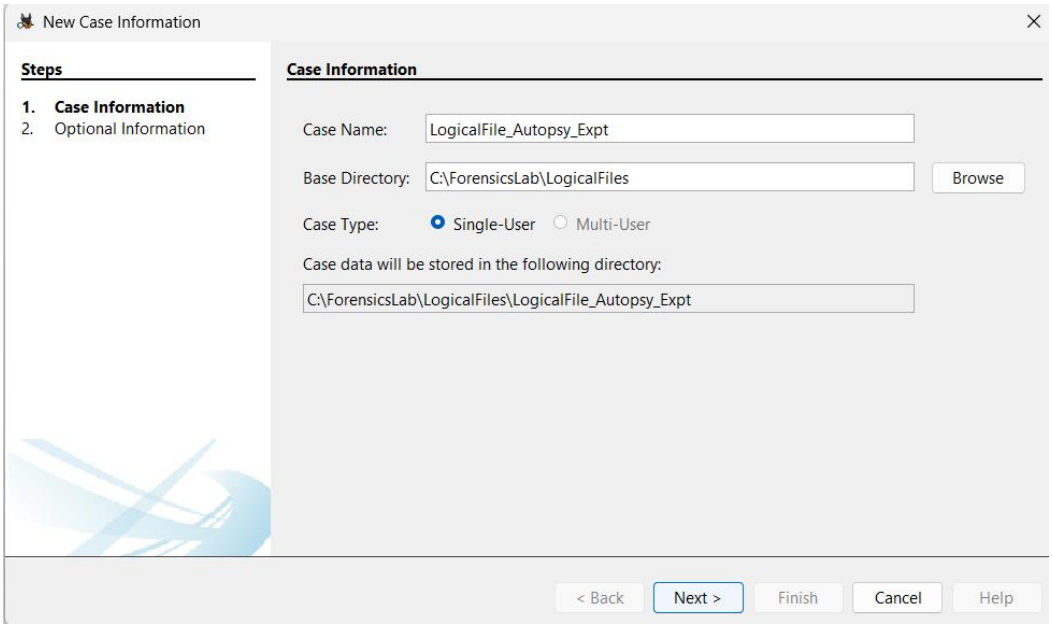
Step 1: Open Autopsy

1. Download and install Autopsy from its official website.
2. Launch the Autopsy application.
3. Select **New Case** from the welcome screen.



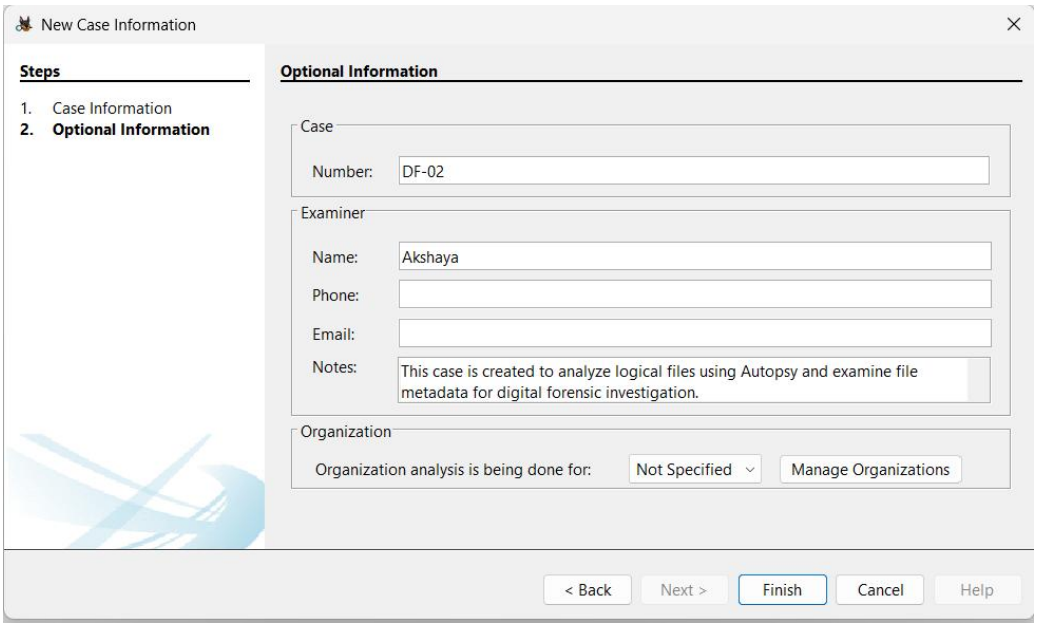
Step 2: Create a new case

1. Enter a suitable case name.
2. Specify the base directory where the case data will be stored.
3. Click **Next** to proceed.



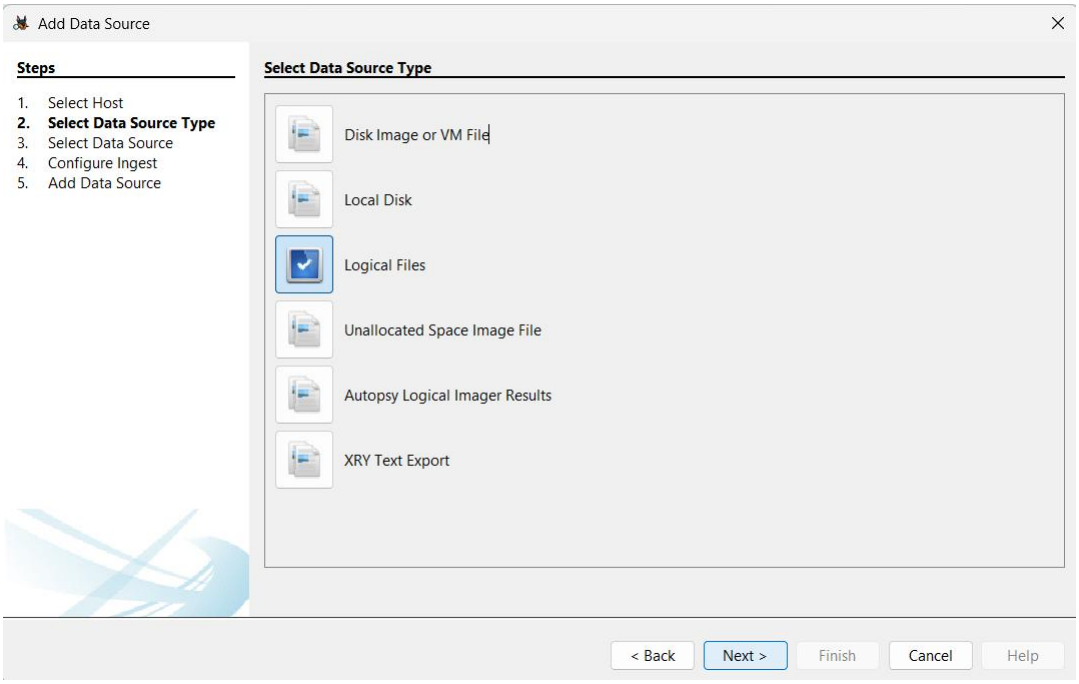
Step 3: Enter case information

1. Enter the case number.
2. Provide the examiner's details.
3. Enter any additional information required by the application.
4. Click **Finish** to create the case.



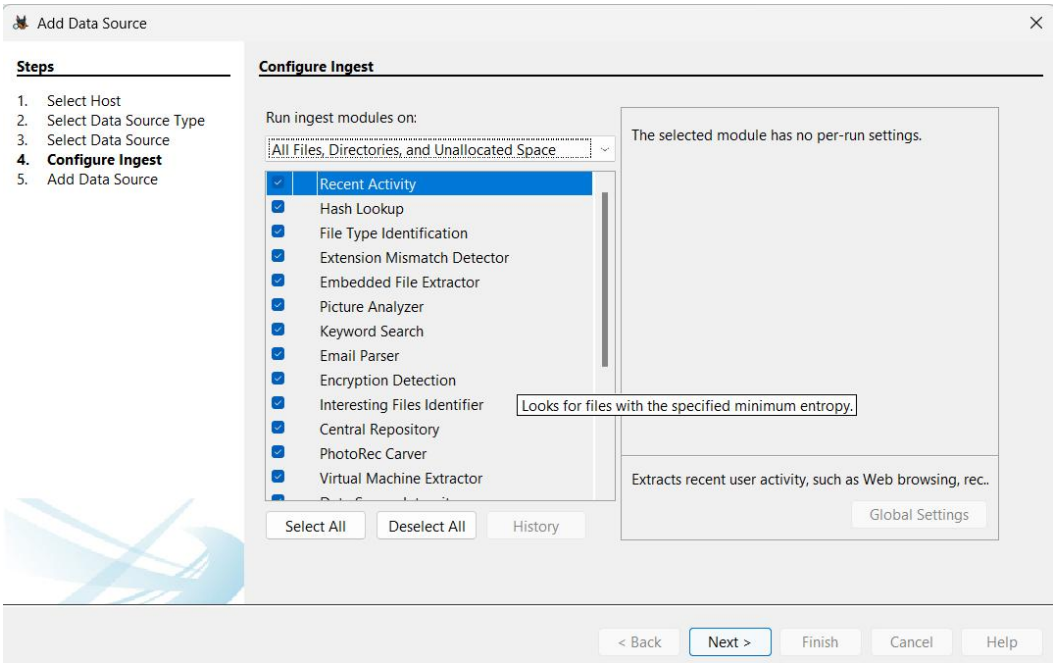
Step 4: Add a data source

1. After creating the case, select the option to add a data source.
2. Choose the appropriate source type, such as a disk image or virtual machine file.
3. Browse to the location of the forensic image.
4. Select the image and proceed to the next step.



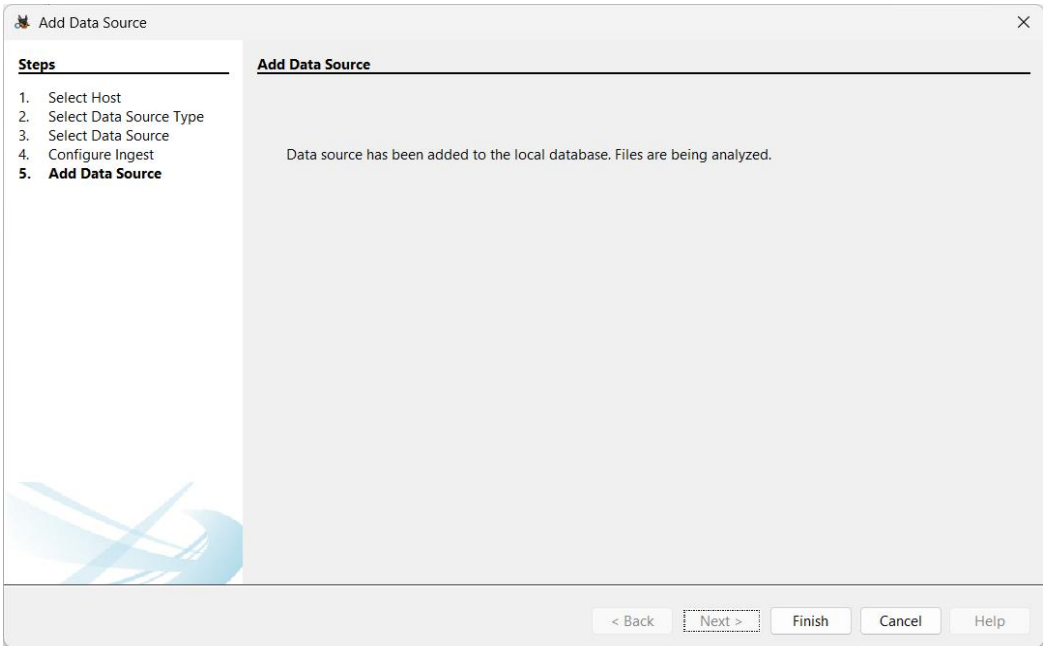
Step 5: Configure ingest modules

1. Review the available ingest modules.
2. Select the modules required for the investigation.
3. Click **Next** to continue.
4. Review the settings and complete the data-source addition.



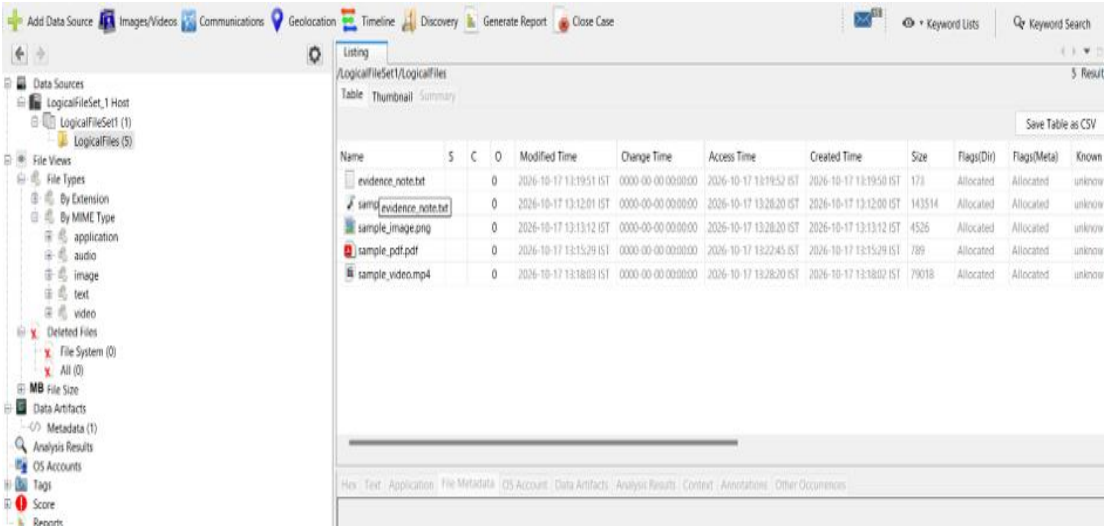
Step 6: Process the data source

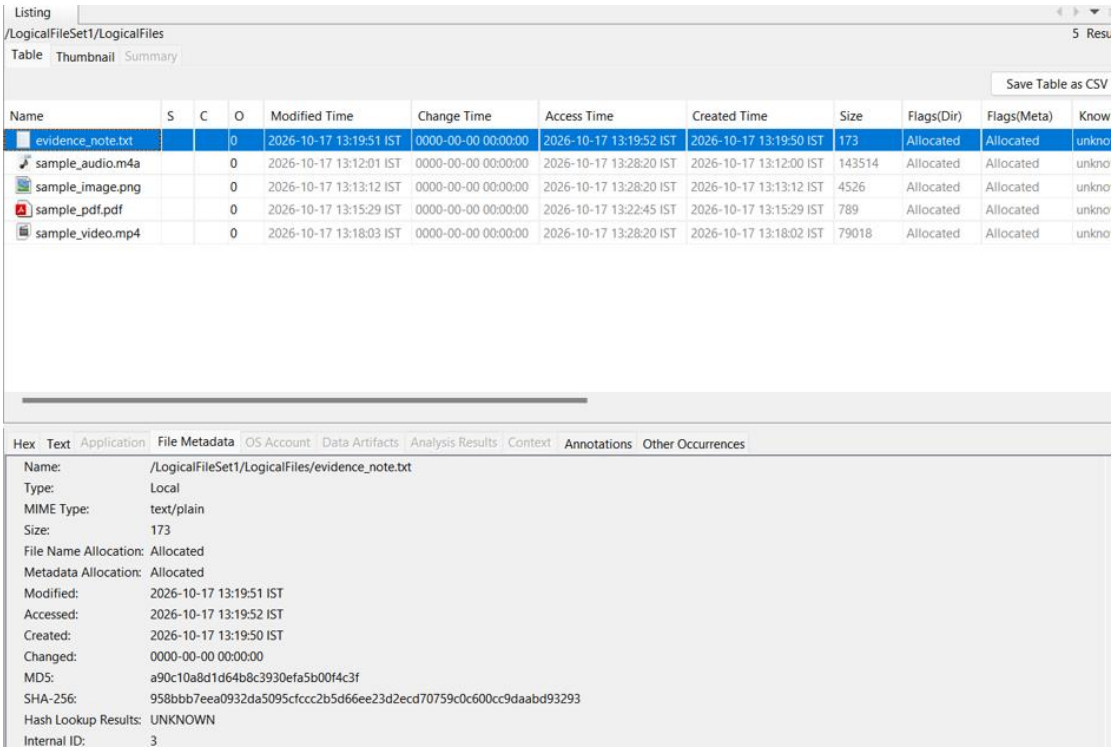
1. Complete the Add Data Source wizard.
2. Allow Autopsy to process the selected image.
3. Wait for processing to finish.
4. Examine the results displayed in the case interface.



Step 7: Examine the forensic artifacts

After processing, examine the available artifacts in Autopsy. Depending on the evidence and the modules selected.





VERIFICATION OF EVIDENCE INTEGRITY USING SHA-256:

After processing the logical files in Autopsy, the integrity of the evidence was cross-verified using the SHA-256 hashing algorithm in Windows Command Prompt.



OBSERVATION:

Autopsy was used to create a forensic case and add a data source for investigation. The selected ingest modules processed the evidence and generated artifacts for examination.

The resulting interface provided access to the files and other information identified during processing. The available artifacts depended on the contents of the selected evidence image.

RESULT:

Thus, the procedure for conducting a digital forensic investigation using Autopsy was studied and performed by creating a case, adding a data source, processing the evidence, and examining the available forensic artifacts.

EXPERIMENT 3

RECOVER DELETED FILES USING FORENSIC TOOLS

AIM:

To recover deleted files from a forensic image using digital forensic tools and verify the recovered data.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- FTK Imager
- Recover My Files
- USB storage device or suitable test evidence source

THEORY:

File deletion does not always immediately erase the underlying data from a storage device. In many file systems, deletion marks the space occupied by a file as available for reuse. Until the data is overwritten, some or all of the deleted file's contents may remain recoverable.

Digital forensic tools help investigators examine storage devices and forensic images to locate and recover deleted files. FTK Imager is used to acquire or examine forensic images, while file recovery tools can search the image for deleted files and other recoverable data.

A forensic image preserves a representation of the source at the time of acquisition, allowing investigators to examine the acquired data without repeatedly accessing the original device.

PROCEDURE:

Step 1: Create a sample file

1. Open Notepad.

2. Type some sample text for testing.
3. Save the file as deleted_secret.txt on a test USB drive or another location.

```
CONFIDENTIAL – FORENSIC EXPERIMENT FILE

Case Reference: CF-2026-001
Project: Cyber and Digital Forensics
Prepared By: Student Name
Date: 17-10-2026

Subject: Sample confidential project document

This file contains simulated evidence for the Autopsy and FTK Imager
laboratory experiments.

Employee ID: EMP-10293
Username: student_forensics
Email: student.forensics@example.com
Project Code: CYBER-DFIR-2026
Access Level: Restricted

Confidential project notes:
The internal audit identified unauthorized access attempts on the
company research server. The suspect may have transferred project
documents to a USB storage device.

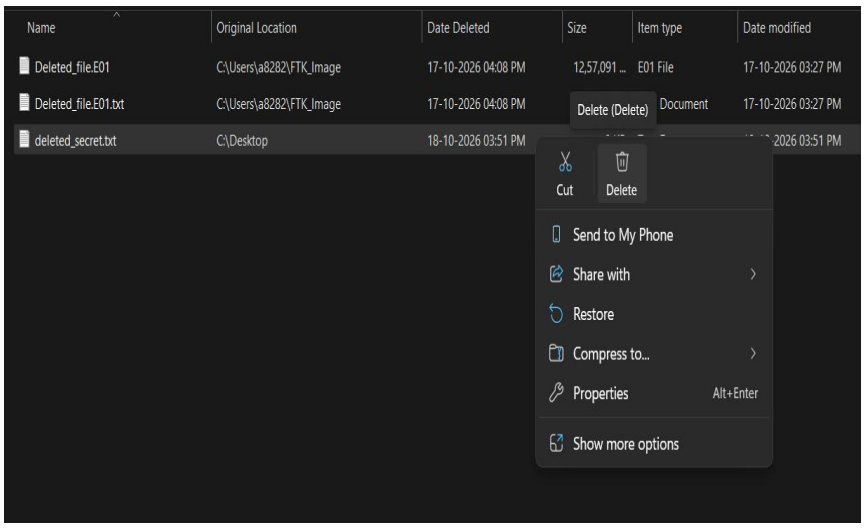
Important keywords:
confidential
unauthorized access
USB transfer
project documents

Password Hint: ForensicsLab@2026

Disclaimer:
This file contains fictional information created only for educational
and forensic analysis purposes.
```

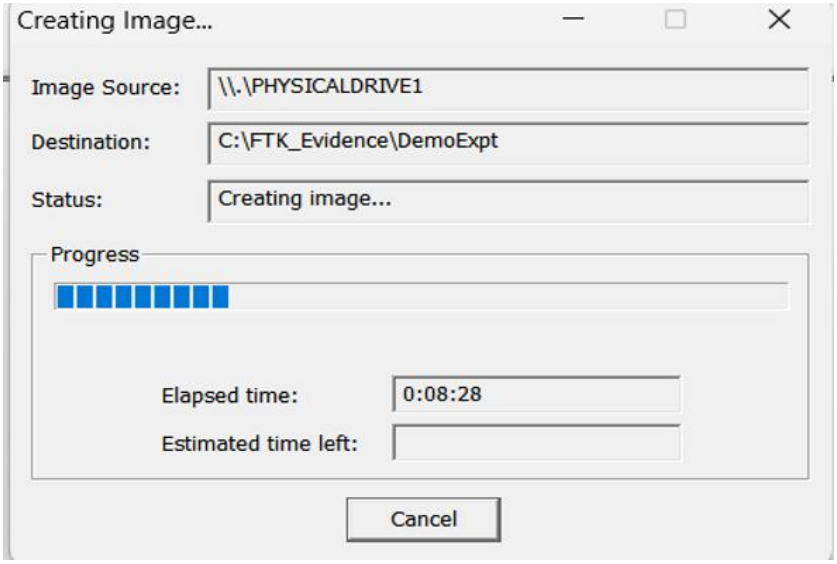
Step 2: Delete the sample file

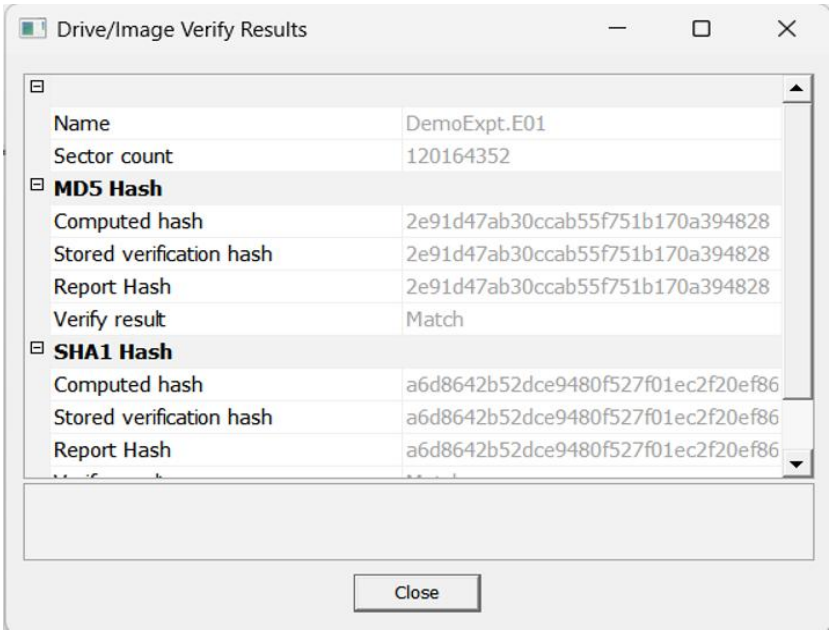
1. Locate deleted_secret.txt.
2. Right-click the file and select **Delete**.
3. Confirm that the file is no longer visible in its original location.

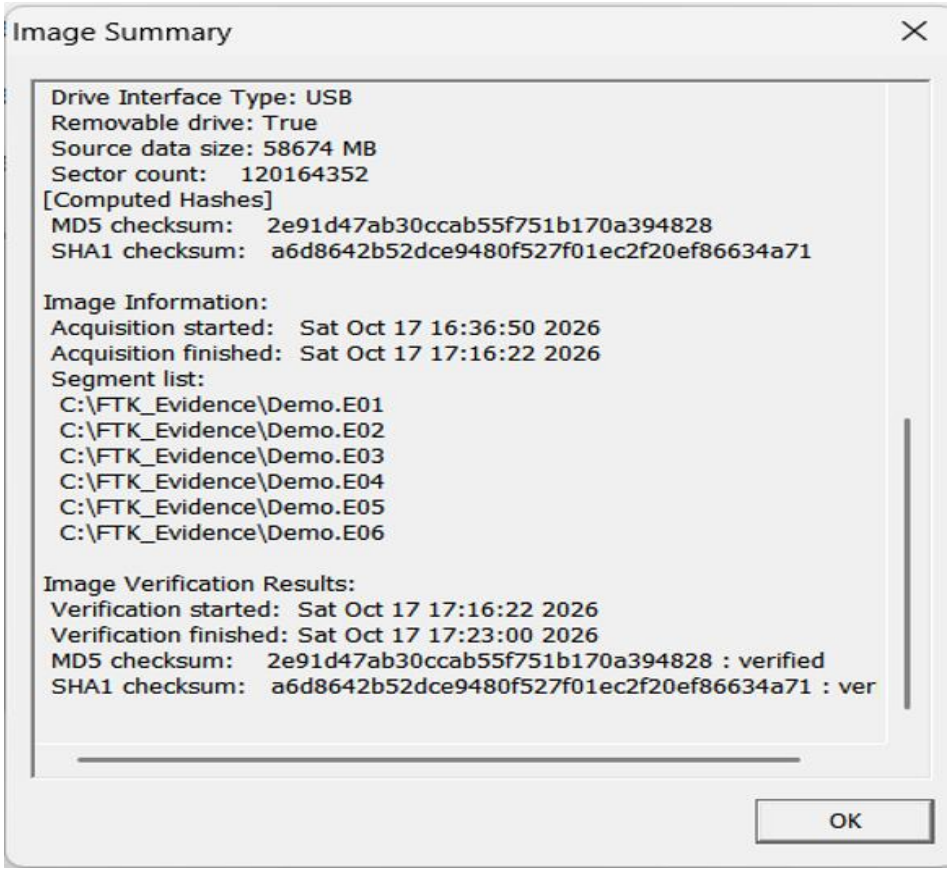


Step 3: Create a forensic image using FTK Imager

1. Open FTK Imager.
2. Select **File → Create Disk Image**.
3. Choose the appropriate source type.
4. Select the test device containing the deleted file.
5. Enter the required case and evidence details.
6. Specify the destination folder and image format.
7. Start the imaging process and wait for it to complete.

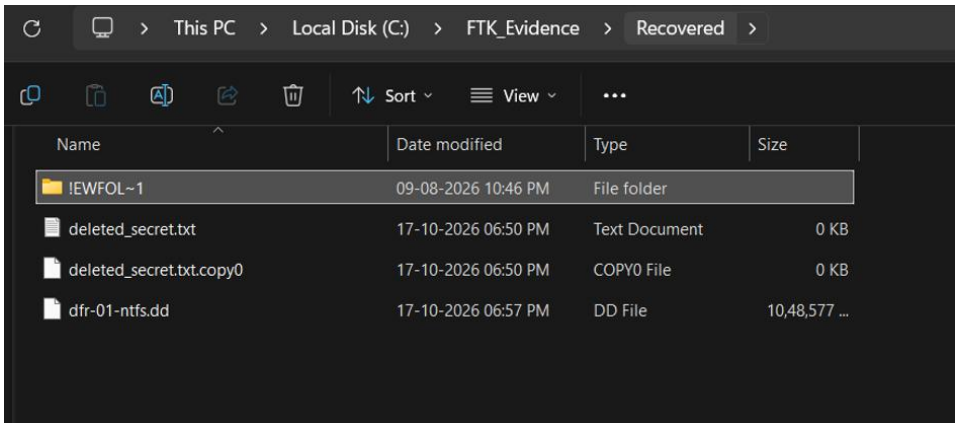






Step 4: Verify the recovered file

1. Open the recovered file.
2. Check whether the original sample text is present.
3. Compare the recovered contents with the text entered before deletion.



OBSERVATION:

A sample text file was created and deleted from a test storage location. A forensic image was then created using FTK Imager and examined using a file recovery tool.

The recovery results were inspected to determine whether the deleted file and its contents could be retrieved. Recovery depended on the file system, the imaging method, and whether the original data had been overwritten.

RESULT:

Thus, the procedure for recovering deleted files using forensic tools was studied and performed. The forensic image was examined for deleted data, and the recovery results were verified where the file was successfully recovered.

EXPERIMENT 4

FIND LAST CONNECTED USB ON YOUR SYSTEM (USB FORENSICS)

AIM:

To identify USB devices previously connected to the computer and examine their details using the **USBDeview** forensic utility.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- Tool: USBDeview
- Computer with a history of USB device connections

THEORY:

USB forensics involves examining records of USB devices connected to a computer. These records can help investigators identify previously connected devices and determine relevant device information.

USBDeview is a Windows utility that lists currently connected USB devices and devices previously used on the system. It can display information such as:

- Device name and description
- Device type
- Serial number, when available
- Date and time the device was connected or added to the system
- Vendor ID (VID)
- Product ID (PID)

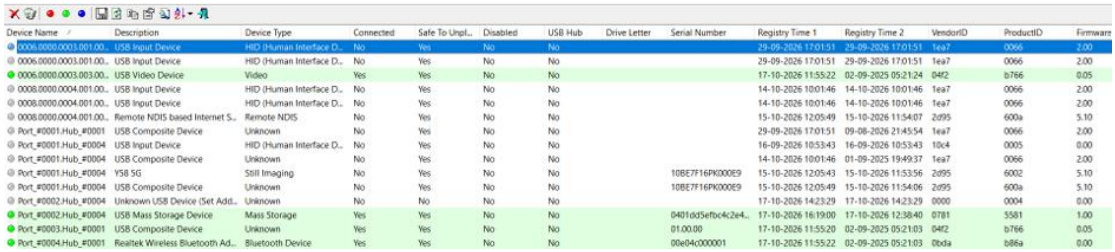
This information can help investigators examine USB connection history during a digital forensic investigation.

PROCEDURE:

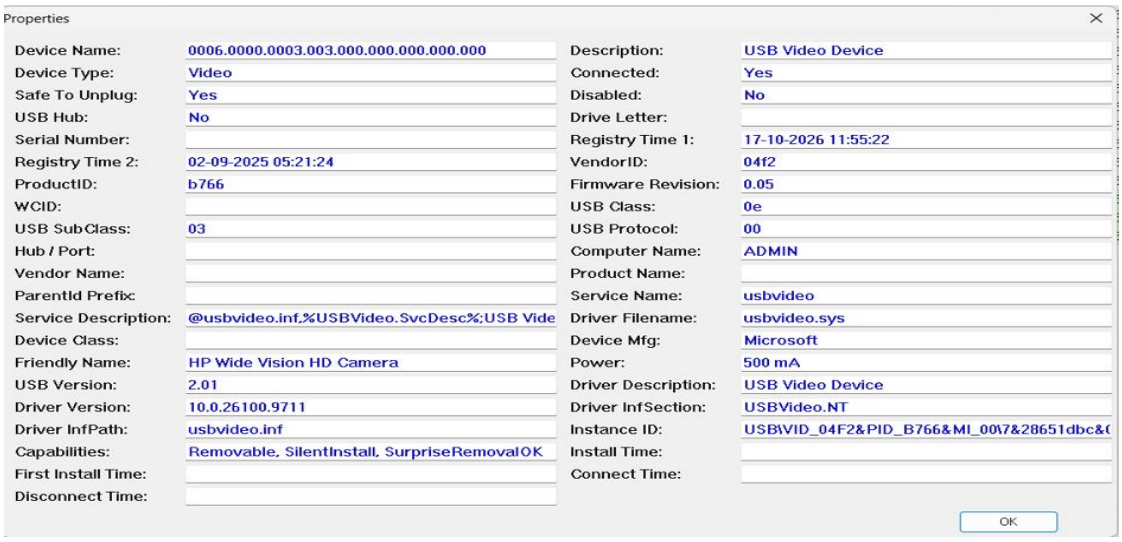
1. Download and launch the USBDeview utility on the Windows system.
2. Allow the utility to display the USB devices detected on the computer.
3. Examine the list of currently connected and previously connected USB devices.
4. Identify the device name and description of the required USB device.
5. Examine the available serial number, Vendor ID, and Product ID.
6. Check the displayed date and time information associated with the device.
7. Record the relevant details for forensic examination.
8. Capture screenshots showing the device list and the selected device's details.

OBSERVATION:

USBDeview displays information about USB devices connected to the system, including devices that were previously used. The available device details can be examined to identify relevant USB connection records.



Device Name	Description	Device Type	Connected	Safe To Unpl...	Disabled	USB Hub	Drive Letter	Serial Number	Registry Time 1	Registry Time 2	VendorID	ProductID	Firmware
0006.0000.0003.001.00...	USB Input Device	HID (Human Interface D...	No	Yes	No	No			29-09-2026 17:01:51	29-09-2026 17:01:51	1ea7	0066	2.00
0006.0000.0003.001.00...	USB Input Device	HID (Human Interface D...	No	Yes	No	No			29-09-2026 17:01:51	29-09-2026 17:01:51	1ea7	0066	2.00
0006.0000.0003.003.00...	USB Video Device	Video	Yes	Yes	No	No			17-10-2026 11:55:22	02-09-2025 05:21:24	04f2	b766	0.05
0008.0000.0004.001.00...	USB Input Device	HID (Human Interface D...	No	Yes	No	No			14-10-2026 10:01:46	14-10-2026 10:01:46	1ea7	0066	2.00
0008.0000.0004.001.00...	USB Input Device	HID (Human Interface D...	No	Yes	No	No			14-10-2026 10:01:46	14-10-2026 10:01:46	1ea7	0066	2.00
0008.0000.0004.001.00...	Remote NDIS based Internet S...	Remote NDIS	No	Yes	No	No			15-10-2026 12:05:49	15-10-2026 11:54:07	2d95	600a	5.10
Port_#0001.Hub_#0001	USB Composite Device	Unknown	No	Yes	No	No			29-09-2026 17:01:51	09-08-2026 21:45:54	1ea7	0066	2.00
Port_#0001.Hub_#0004	USB Input Device	HID (Human Interface D...	No	Yes	No	No			16-09-2026 10:53:43	16-09-2026 10:53:43	10c4	0005	0.00
Port_#0001.Hub_#0004	USB Composite Device	Unknown	No	Yes	No	No			14-10-2026 10:01:46	01-09-2025 19:49:37	1ea7	0066	2.00
Port_#0001.Hub_#0004	Y58 5G	Still Imaging	No	Yes	No	No		10BE7F16PK000E9	15-10-2026 12:05:43	15-10-2026 11:53:56	2d95	6002	5.10
Port_#0001.Hub_#0004	USB Composite Device	Unknown	No	Yes	No	No		10BE7F16PK000E9	15-10-2026 12:05:49	15-10-2026 11:54:06	2d95	600a	5.10
Port_#0002.Hub_#0004	Unknown USB Device (Set Add...	Unknown	No	No	No	No			17-10-2026 14:23:29	17-10-2026 14:23:29	0000	0004	0.00
Port_#0002.Hub_#0004	USB Mass Storage Device	Mass Storage	Yes	Yes	No	No		0401dd5efbc4c2e4...	17-10-2026 16:19:00	17-10-2026 12:38:40	0781	5581	1.00
Port_#0003.Hub_#0001	USB Composite Device	Unknown	Yes	Yes	No	No		01.00.00	17-10-2026 11:55:20	02-09-2025 05:21:03	04f2	b766	0.05
Port_#0004.Hub_#0001	Realtek Wireless Bluetooth Ad...	Bluetooth Device	Yes	Yes	No	No		00e04c000001	17-10-2026 11:55:22	02-09-2025 05:21:03	0bda	b86a	0.00



Properties

Device Name:	0006.0000.0003.003.000.000.000.000.000	Description:	USB Video Device
Device Type:	Video	Connected:	Yes
Safe To Unplug:	Yes	Disabled:	No
USB Hub:	No	Drive Letter:	
Serial Number:		Registry Time 1:	17-10-2026 11:55:22
Registry Time 2:	02-09-2025 05:21:24	VendorID:	04f2
ProductID:	b766	Firmware Revision:	0.05
WCID:		USB Class:	0e
USB SubClass:	03	USB Protocol:	00
Hub / Port:		Computer Name:	ADMIN
Vendor Name:		Product Name:	
ParentId Prefix:		Service Name:	usbvideo
Service Description:	@usbvideo.inf,%USBVideo.SvcDesc%;USB Vide	Driver Filename:	usbvideo.sys
Device Class:		Device Mfg:	Microsoft
Friendly Name:	HP Wide Vision HD Camera	Power:	500 mA
USB Version:	2.01	Driver Description:	USB Video Device
Driver Version:	10.0.26100.9711	Driver InfSection:	USBVideo.NT
Driver InfPath:	usbvideo.inf	Instance ID:	USB\VID_04F2&PID_B766&MI_00\7&28651dbc&(
Capabilities:	Removable, SilentInstall, SurpriseRemovalOK	Install Time:	
First Install Time:		Connect Time:	
Disconnect Time:			

OK

RESULT:

Thus, USBDeview was used to examine USB device connection records and view the available device details for forensic investigation.

EXPERIMENT 5

VIEW LAST ACTIVITY OF YOUR PC

AIM:

To examine the recent activities and system events recorded on a Windows computer using the **LastActivityView** forensic utility.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- Tool: LastActivityView

THEORY:

LastActivityView is a utility for Windows that collects information from different sources on the system and displays a log of user activities and events that occurred on the computer.

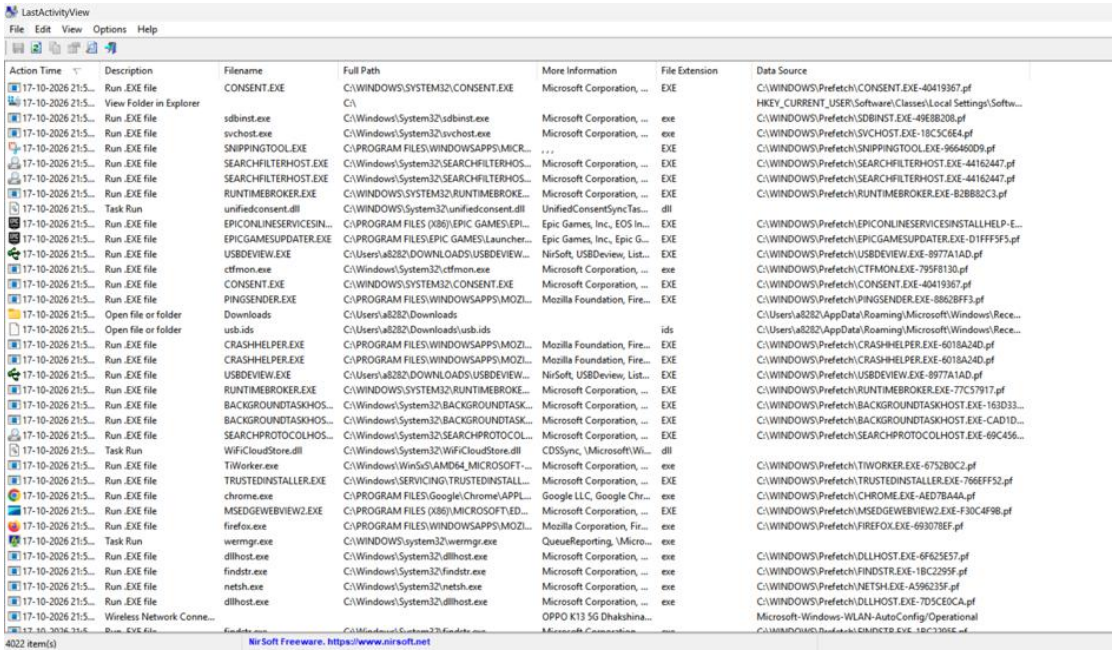
It can help investigators examine activity records, such as system events and certain recent user actions. The information available depends on the records present on the computer.

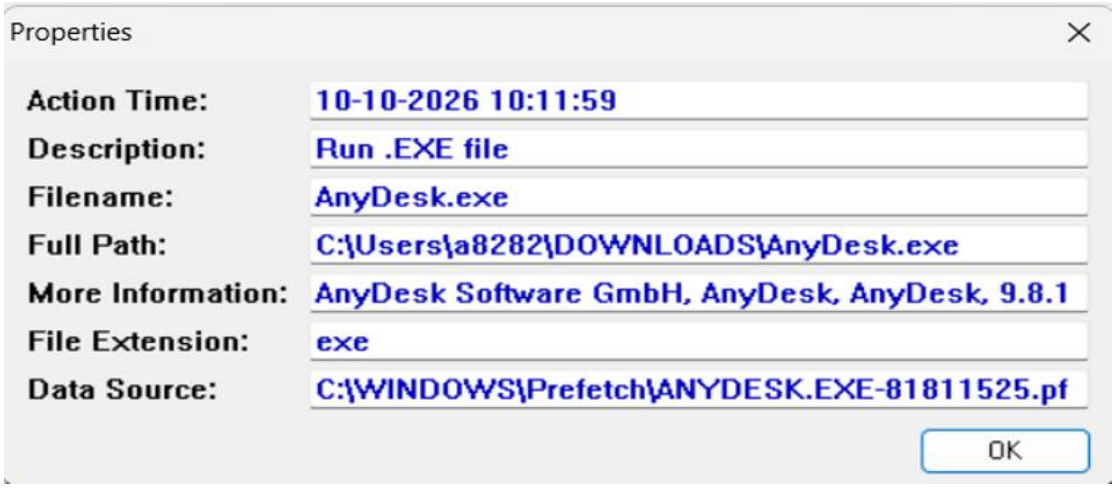
PROCEDURE:

1. Download and launch the LastActivityView utility on the Windows computer.
2. Allow the utility to scan the available system records.
3. Examine the activity log displayed in the main window.
4. Observe the available details, such as the activity description, filename or application, and recorded date and time, where available.
5. Identify relevant entries for recent computer activities.
6. Record the observations for forensic analysis.
7. Capture screenshots of the activity log and relevant entries.

OBSERVATION:

LastActivityView displays available records of activities and events associated with the Windows computer. These records can be examined to understand certain aspects of the system's recent activity.





RESULT:

Thus, LastActivityView was used to examine available records of recent computer activities and system events for digital forensic investigation.

EXPERIMENT 6

EXTRACTION OF BROWSER ARTIFACTS

AIM

To extract and examine browser artifacts, including cached files and browsing history, using ChromeCacheView and ChromeHistoryView.

SOFTWARE REQUIREMENTS

- Operating System: Windows
- Web Browser: Google Chrome
- Tools: ChromeCacheView and ChromeHistoryView

THEORY

Browser artifacts are data generated and stored by a web browser during internet usage. These artifacts can provide useful information during digital forensic investigations.

ChromecacheView is a utility that reads the cache folder of Google Chrome and displays information about cached files. The information may include the URL, file type, cache filename, file size, and access time, depending on the available records.

ChromeHistoryView is a utility that reads the history data of Google Chrome and displays records of websites visited by the user. It can show information such as the URL, page title, visit count, and last visit time, depending on the available records.

These tools help investigators examine browser cache and browsing history as part of a digital forensic investigation.

4. PROCEDURE

A. EXAMINATION USING CHROMECACHEVIEW

1. Download and launch ChromeCacheView on the Windows computer.
2. Allow the utility to locate and read the Google Chrome cache data.

3. Examine the list of cached files displayed in the main window.
4. Observe the available details, including the URL, file type, cache filename, file size, and access time.
5. Identify relevant cached entries for examination.
6. Capture a screenshot showing the cached file records.

B. EXAMINATION USING CHROMEHISTORYVIEW

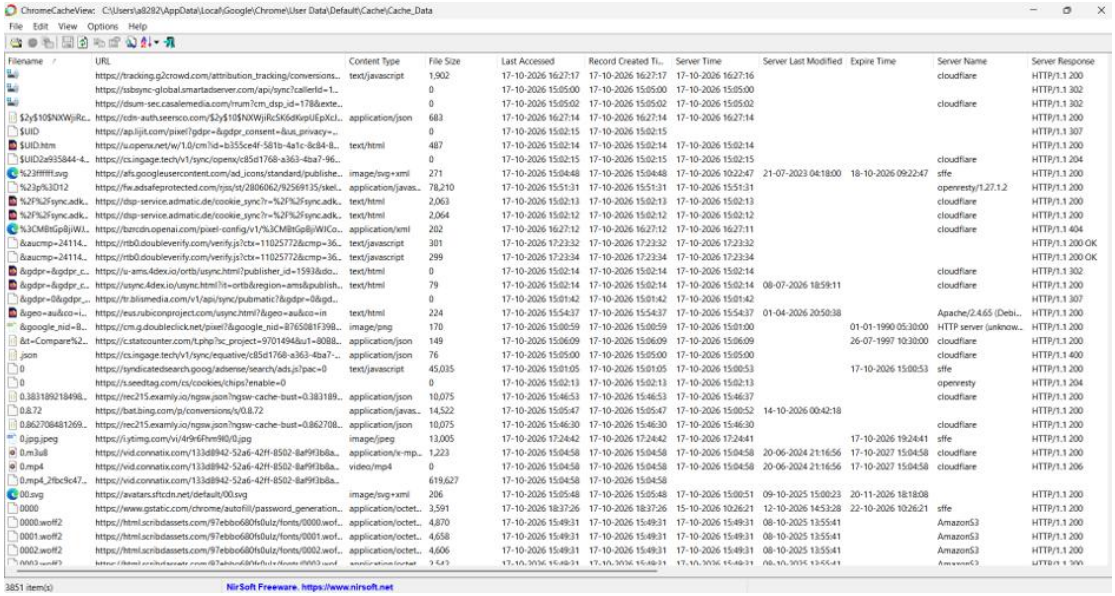
1. Download and launch ChromeHistoryView on the Windows computer.
2. Allow the utility to locate and read the Google Chrome browsing history data.
3. Examine the list of browsing history records displayed in the main window.
4. Observe the available details, including the URL, page title, visit count, and last visit time.
5. Identify relevant browsing history entries.
6. Capture a screenshot showing the extracted browsing history records.

OBSERVATION:

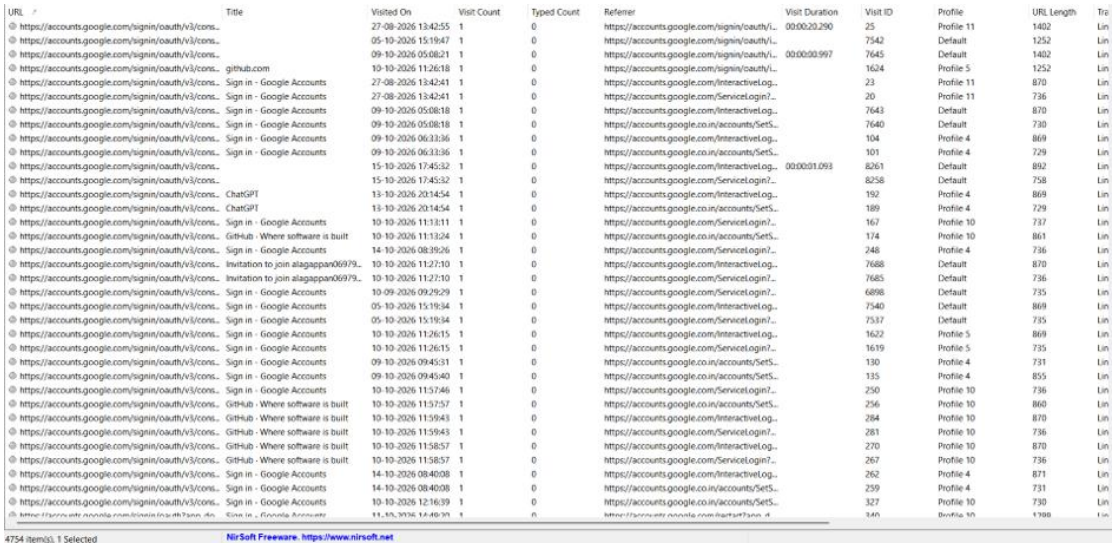
The two tools were used to examine different types of browser artifacts.

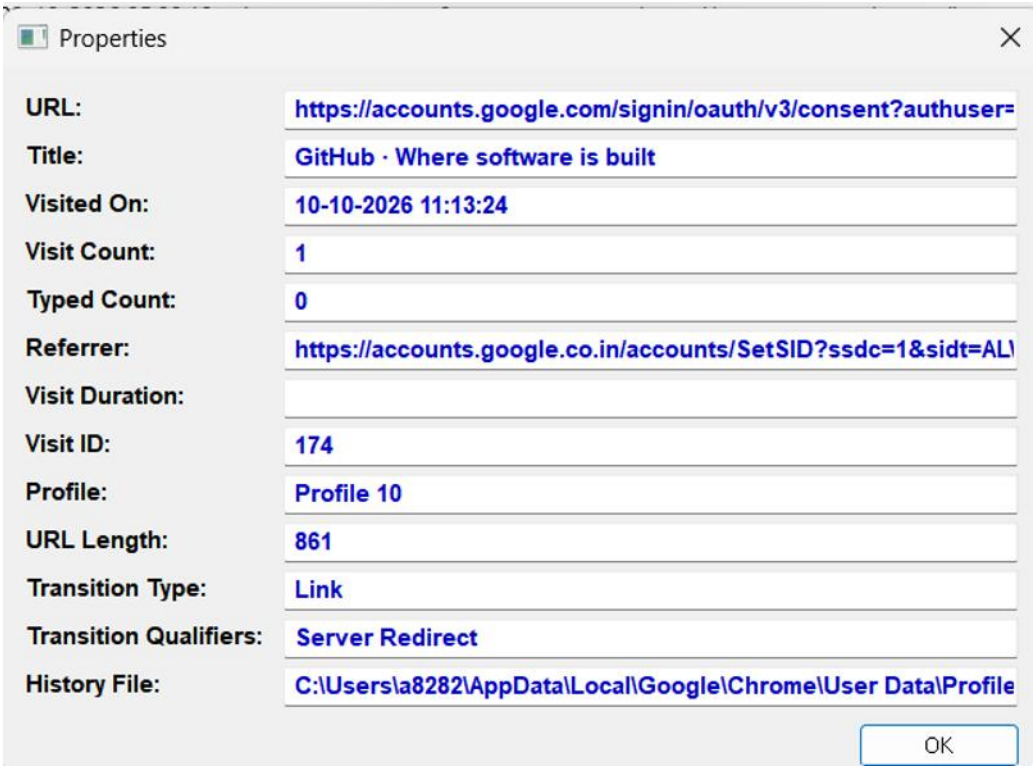
S. NO.	**TOOL NAME**	**ARTIFACT EXAMINED**	**OBSERVATION**
1	ChromeCacheView	Cached files	Cached file details were displayed where available.
2	ChromeHistoryView	Browsing history	Website history records were displayed where available.

DETAILS OBSERVED USING CHROMECACHEVIEW



DETAILS OBSERVED USING CHROMEHISTORYVIEW





RESULT:

Thus, browser artifacts were examined using ChromeCacheView and ChromeHistoryView. Cached file information and available browsing history records were extracted and analyzed for digital forensic investigation.

EXPERIMENT 7

COMPARISON OF TWO FILES FOR FORENSIC INVESTIGATION USING COMPARE IT SOFTWARE

AIM:

To compare two files using Compare It software and identify differences between their contents for digital forensic investigation.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- Software: Compare It
- Two sample files for comparison

THEORY:

File comparison is a technique used in digital forensics to identify differences between two files. It helps investigators determine whether a file has been modified or whether two files contain different information.

Compare It is a file comparison tool that displays two files side by side and highlights the differences using colours. It helps users identify added, deleted, or modified content between files.

The comparison results can assist investigators in examining changes in documents and understanding possible file modifications.

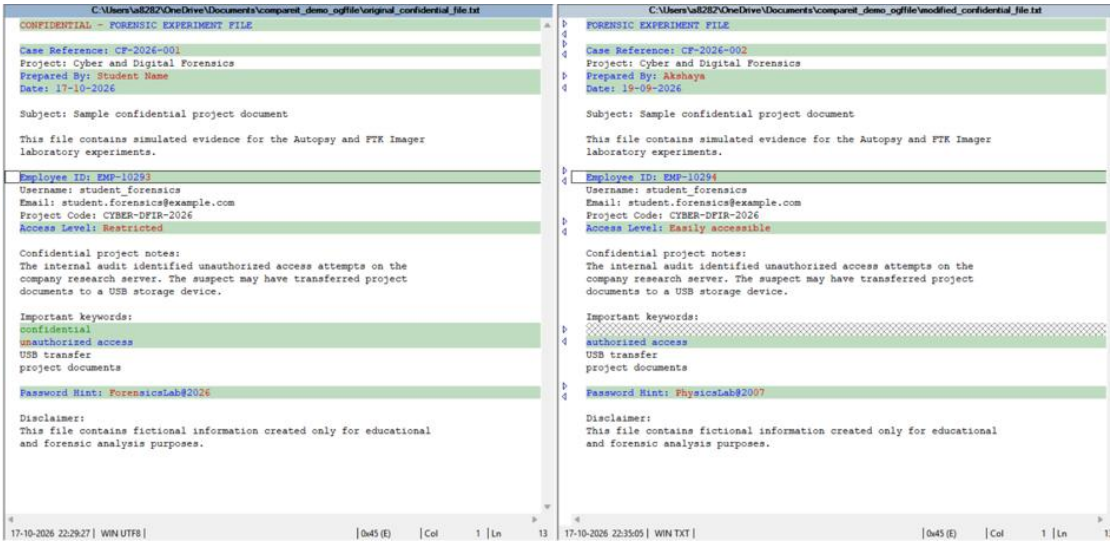
PROCEDURE:

1. Download and install Compare It software on the Windows computer.
2. Prepare two sample files for comparison. The files may contain similar text with a few differences.
3. Launch the Compare It application.
4. Select the first file and click **OPEN**.
5. Select the second file and click **OPEN**.

6. Allow the software to display both files side by side.
7. Examine the highlighted sections to identify differences between the files.
8. Observe any added, deleted, or modified text.
9. Record the observations and capture a screenshot of the comparison results.

OBSERVATION:

Compare It displays both files side by side and highlights differences in their contents.





RESULT:

Thus, two files were compared using Compare It software, and the differences in their contents were identified for forensic examination.

EXPERIMENT 8

COLLECTING EMAIL EVIDENCE IN VICTIM PC

AIM:

To collect email-related evidence from a computer by acquiring its RAM using DumpIt and examining the acquired memory for relevant information.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- Tool: DumpIt
- Memory analysis tool or scripts specified in the lab procedure

THEORY:

Email forensics involves identifying and examining email-related information available on a computer. Such information may help investigators understand communication activities during a digital forensic investigation.

DUMPIT is a memory acquisition utility used to create a physical memory dump of a Windows computer. Volatile memory may contain temporary information associated with running applications, including browser-based email activity.

After acquiring the memory dump, investigators can examine it using suitable forensic analysis tools to identify potentially relevant artifacts, such as email addresses and other email-related information.

The acquired memory must be preserved carefully to maintain the integrity of the evidence.

PROCEDURE:

A. CAPTURING RAM USING DUMPIT

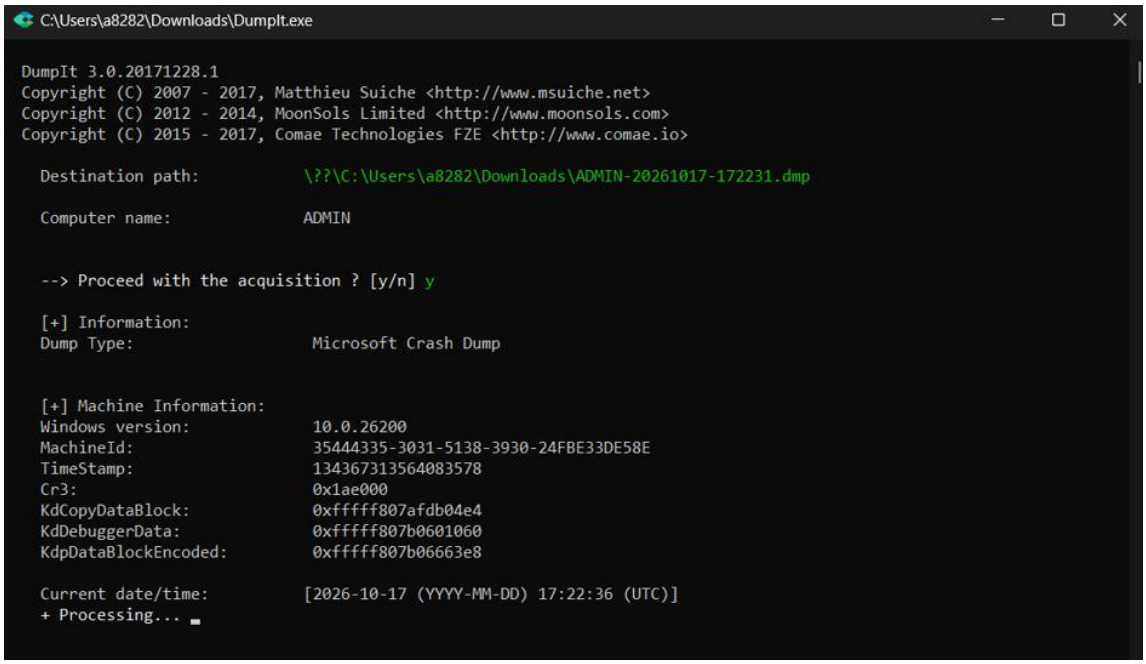
1. Download and prepare the DumpIt utility on the Windows computer.
2. Run DumpIt.exe with the required administrative permissions.

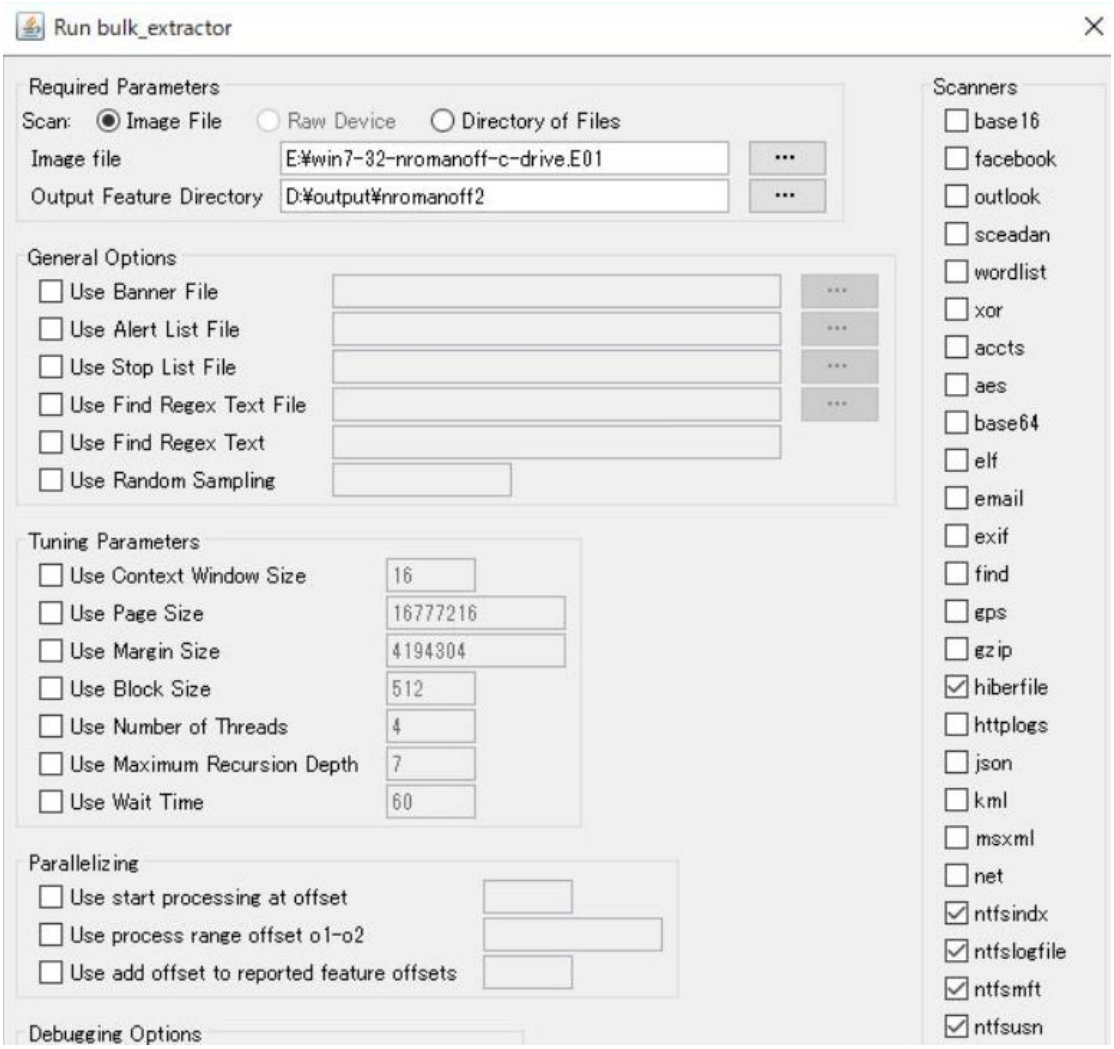
3. Read the confirmation prompt displayed by the tool.
4. Enter Y to confirm and begin the memory acquisition, as instructed by the manual.
5. Wait for the acquisition process to complete.
6. Locate the generated raw memory dump in the directory specified by the tool.
7. Record the dump filename and file size.

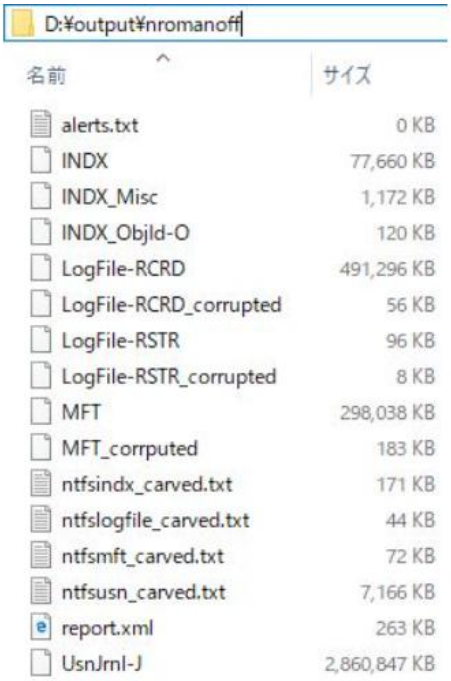
B. EXAMINING EMAIL-RELATED EVIDENCE

1. Open the acquired memory dump using the memory examination procedure specified in the lab manual.
2. Examine the available results for email-related information.
3. Open email.txt to inspect the email-related information identified by the procedure.
4. Open email_histogram.txt to examine the email-related summary or histogram output.
5. Record relevant observations from the generated files.
6. Preserve the acquired dump and analysis results for documentation.

OBSERVATION:




Run bulk_extractor
Required Parameters
Scan: Image File Raw Device Directory of Files
Image file E:¥win7-32-nromanoff-c-drive.E01
Output Feature Directory D:¥output¥nromanoff2
General Options
Use Banner File
Use Alert List File
Use Stop List File
Use Find Regex Text File
Use Find Regex Text
Use Random Sampling
Tuning Parameters
Use Context Window Size 16
Use Page Size 16777216
Use Margin Size 4194304
Use Block Size 512
Use Number of Threads 4
Use Maximum Recursion Depth 7
Use Wait Time 60
Parallelizing
Use start processing at offset
Use process range offset o1-o2
Use add offset to reported feature offsets
Debugging Options
Scanners
base16
facebook
outlook
sceadan
wordlist
xor
accts
aes
base64
elf
email
exif
find
gps
gzip
hiberfile
httplogs
json
kml
msxml
net
ntfsindx
ntfslogfile
ntfsmft
ntfsusn


D:¥output¥nromanoff
名前
サイズ
alerts.txt 0 KB
INDX 77,660 KB
INDX_Misc 1,172 KB
INDX_ObjId-O 120 KB
LogFile-RCRD 491,296 KB
LogFile-RCRD_corrupted 56 KB
LogFile-RSTR 96 KB
LogFile-RSTR_corrupted 8 KB
MFT 298,038 KB
MFT_corrputed 183 KB
ntfsindx_carved.txt 171 KB
ntfslogfile_carved.txt 44 KB
ntfsmft_carved.txt 72 KB
ntfsusn_carved.txt 7,166 KB
report.xml 263 KB
UsnJrnl-J 2,860,847 KB

RESULT:

Thus, the procedure for collecting email-related evidence from a computer using DumpIt for RAM acquisition and subsequent memory examination was studied.

EXPERIMENT 9

HIDING AND EXTRACTING A TEXT FILE BEHIND AN IMAGE/AUDIO FILE USING COMMAND PROMPT

AIM:

To demonstrate how to hide a text file behind an image file and an audio file using Command Prompt, with WinRAR used for preparing the archive in the audio method.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- Command Prompt (CMD)
- WinRAR (for the audio file method)
- Image file (B.jpeg)
- Audio file (sound.mp3)
- Text files (A.txt and secret.txt)

THEORY:

File concatenation is a technique in which the contents of two files are combined into a single output file. In this experiment, a text file is appended to an image file using Command Prompt. For the audio method, the text file is first compressed into a ZIP archive using WinRAR, and the archive is then appended to an audio file using Command Prompt.

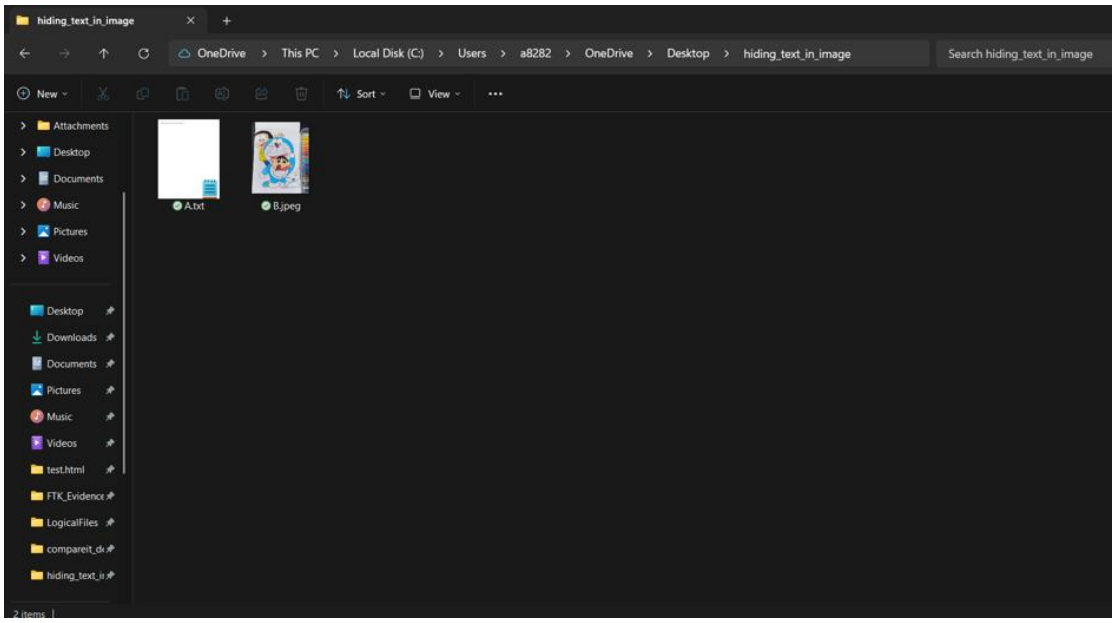
The resulting files contain the original image or audio data followed by the additional data. This technique demonstrates how information can be concealed within a combined file. Digital forensic investigators can examine file structures to identify additional data and investigate whether concealed information can be recovered.

PROCEDURE:

A. HIDING A TEXT FILE BEHIND AN IMAGE FILE USING CMD

1. Create a folder named hiding_text_in_image on the Desktop.

2. Place the image file B.jpeg and text file A.txt inside the folder.
3. Open Command Prompt.
4. Navigate to the working directory using the following command:



Execute the following command to combine the files:

copy /b B.jpeg + A.txt C.jpeg

Where,

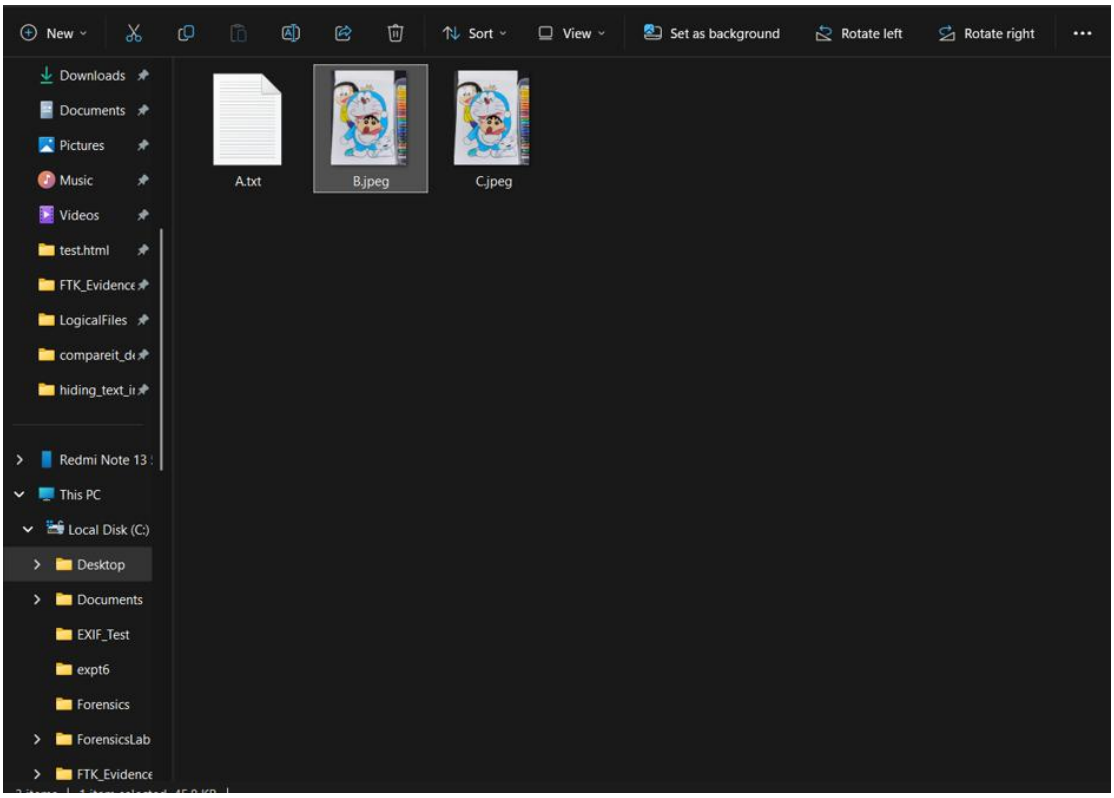
- B.jpeg — Original image file.
- A.txt — Text file to be appended.
- C.jpeg — Combined output file.
- /b — Specifies binary mode.

```
C:\Desktop>cd hiding_text_in_image

C:\Desktop\hiding_text_in_image>copy /b B.jpeg + A.txt C.jpeg
B.jpeg
A.txt
        1 file(s) copied.

C:\Desktop\hiding_text_in_image>
```

1. Wait for the command to complete.
2. Verify that the output file C.jpeg is created in the working directory.
3. Record the output displayed in Command Prompt.





B. HIDING A TEXT FILE BEHIND AN AUDIO FILE USING WINRAR AND CMD

1. Create a text file named secret.txt using Notepad.
2. Enter sample text and save the file.
3. Select secret.txt and use WinRAR to create a ZIP archive named secret.zip.
4. Ensure that sound.mp3 and secret.zip are in the same working directory.
5. Open Command Prompt and navigate to the Steganography folder.

Execute the following command to append the ZIP archive to the audio file:

copy /b sound.mp3+secret.zip newfile.mp3

Where,

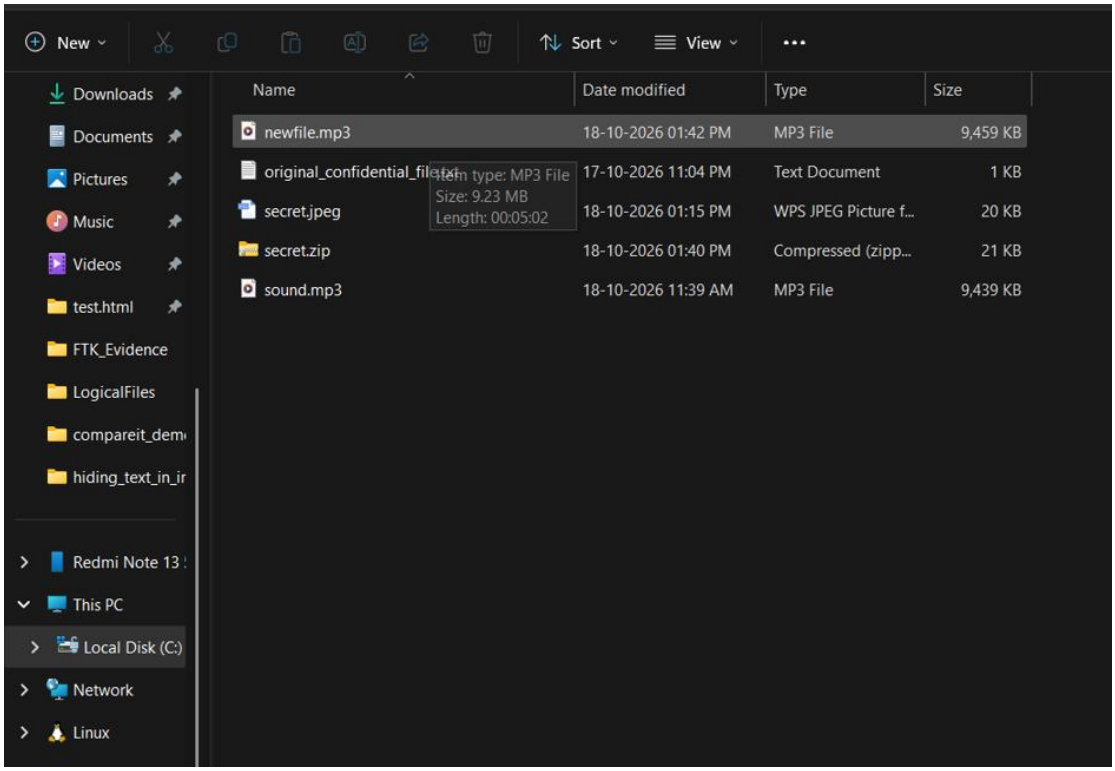
- sound.mp3 — Original audio file.
- secret.zip — ZIP archive containing the text file.
- newfile.mp3 — Combined output file.
- /b — Specifies binary mode.

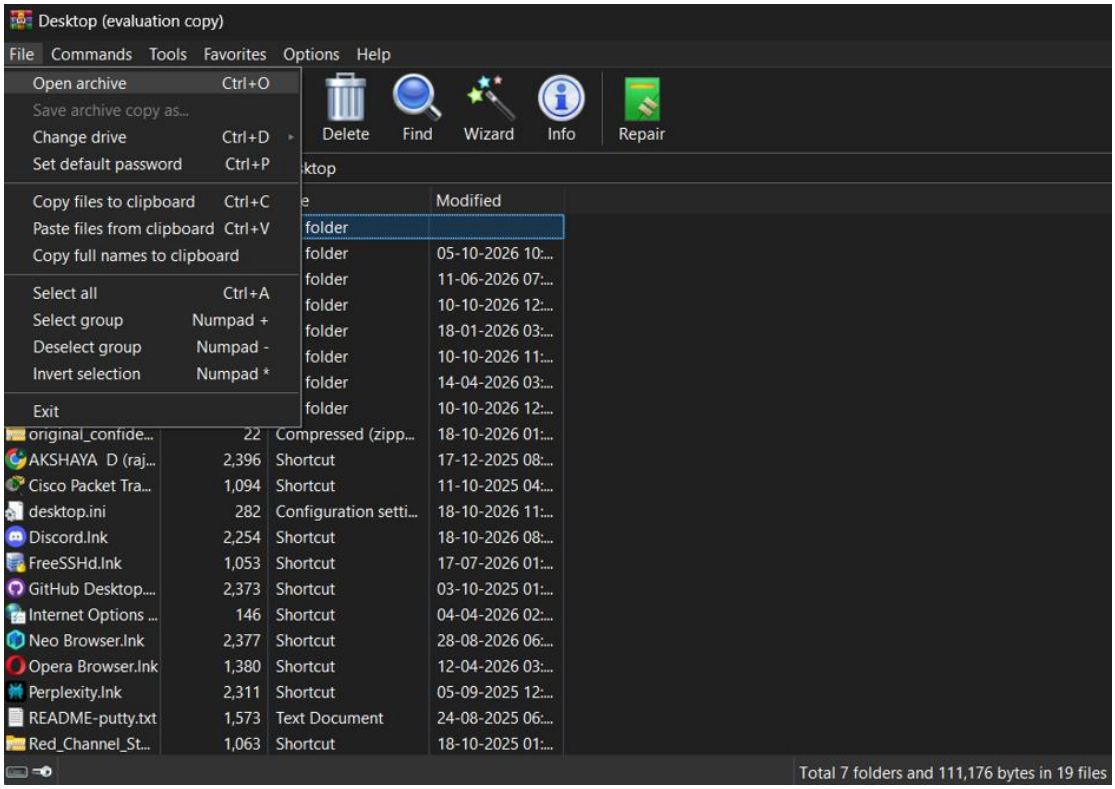
```
C:\Desktop>cd Steganography

C:\Desktop\Steganography>copy /b sound.mp3+secret.zip newfile.mp3
sound.mp3
secret.zip
        1 file(s) copied.

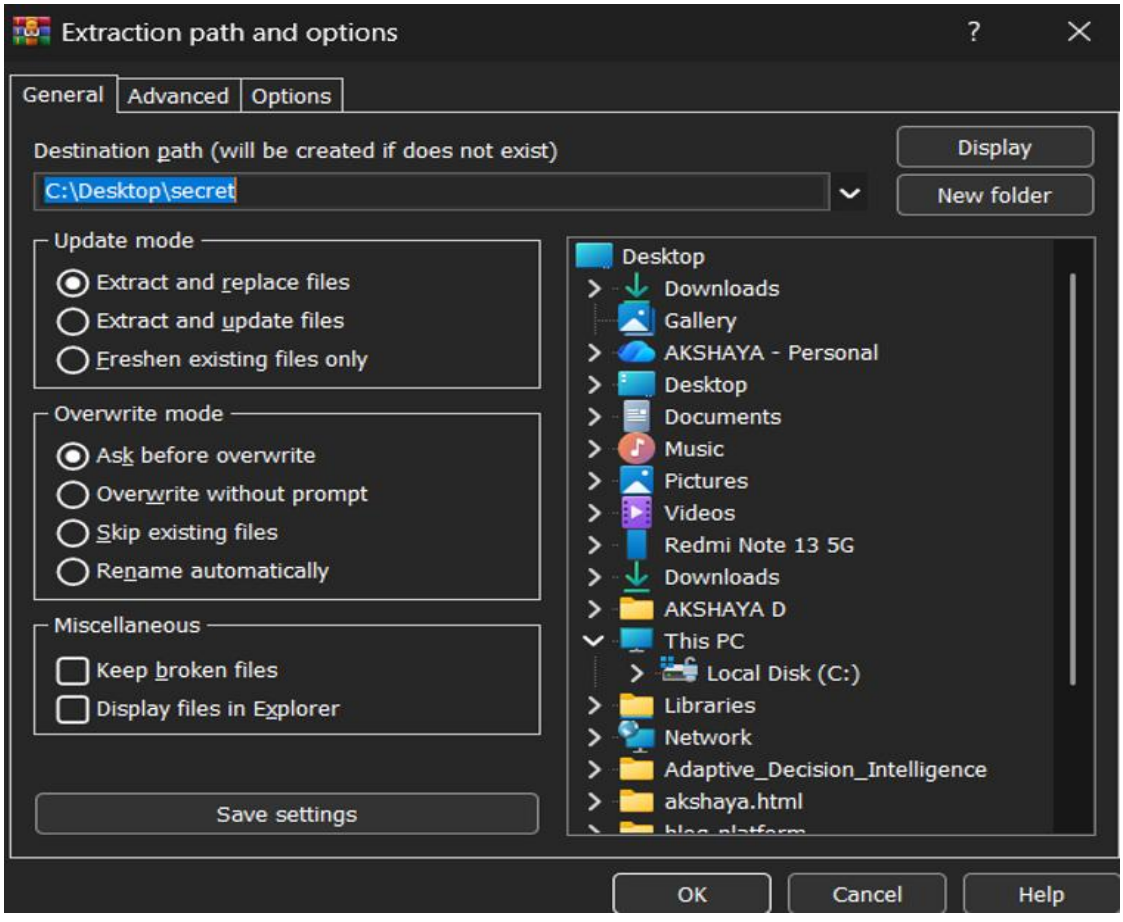
C:\Desktop\Steganography>
```

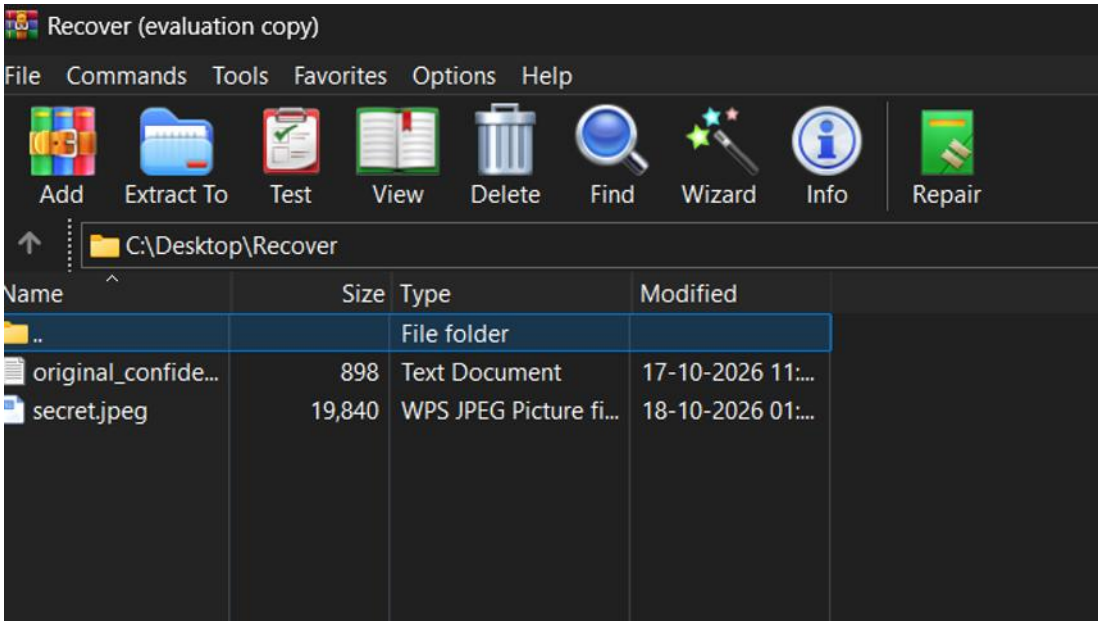
1. Wait for the command to complete.
2. Verify that the output file newfile.mp3 is created.
3. Open newfile.mp3 in Notepad to observe its binary content, if required.
4. To attempt recovery, make a copy of newfile.mp3 and try opening the copy using WinRAR. If the appended archive is recognized, extract its contents and verify whether secret.txt can be recovered.


New
Sort
View
Downloads
Documents
Pictures
Music
Videos
test.html
FTK_Evidence
LogicalFiles
compareit_dem
hiding_text_in_ir
Redmi Note 13
This PC
Local Disk (C:)
Network
Linux
Name
Date modified
Type
Size
newfile.mp3
18-10-2026 01:42 PM
MP3 File
9,459 KB
original_confidential_file.txt
17-10-2026 11:04 PM
Text Document
1 KB
Item type: MP3 File
Size: 9.23 MB
Length: 00:05:02
secret.jpeg
18-10-2026 01:15 PM
WPS JPEG Picture f...
20 KB
secret.zip
18-10-2026 01:40 PM
Compressed (zipp...
21 KB
sound.mp3
18-10-2026 11:39 AM
MP3 File
9,439 KB


Desktop (evaluation copy)
File
Commands
Tools
Favorites
Options
Help
Open archive Ctrl+O
Save archive copy as...
Change drive Ctrl+D
Set default password Ctrl+P
Copy files to clipboard Ctrl+C
Paste files from clipboard Ctrl+V
Copy full names to clipboard
Select all Ctrl+A
Select group Numpad +
Deselect group Numpad -
Invert selection Numpad *
Exit
Delete
Find
Wizard
Info
Repair
Modified
folder
05-10-2026 10:...
11-06-2026 07:...
10-10-2026 12:...
18-01-2026 03:...
10-10-2026 11:...
14-04-2026 03:...
10-10-2026 12:...
original_confide... 22 Compressed (zipp... 18-10-2026 01:...
AKSHAYA D (raj... 2,396 Shortcut 17-12-2025 08:...
Cisco Packet Tra... 1,094 Shortcut 11-10-2025 04:...
desktop.ini 282 Configuration setti... 18-10-2026 11:...
Discord.lnk 2,254 Shortcut 18-10-2026 08:...
FreeSSHd.lnk 1,053 Shortcut 17-07-2026 01:...
GitHub Desktop.... 2,373 Shortcut 03-10-2025 01:...
Internet Options ... 146 Shortcut 04-04-2026 02:...
Neo Browser.lnk 2,377 Shortcut 28-08-2026 06:...
Opera Browser.lnk 1,380 Shortcut 12-04-2026 03:...
Perplexity.lnk 2,311 Shortcut 05-09-2025 12:...
README-putty.txt 1,573 Text Document 24-08-2025 06:...
Red_Channel_St... 1,063 Shortcut 18-10-2025 01:...
Total 7 folders and 111,176 bytes in 19 files


View - newfile.mp3
File Edit View Help
original_confidential_file.txt
secret.jpeg


Extraction path and options
General
Advanced
Options
Destination path (will be created if does not exist)
C:\Desktop\secret
Display
New folder
Update mode
Extract and replace files
Extract and update files
Freshen existing files only
Overwrite mode
Ask before overwrite
Overwrite without prompt
Skip existing files
Rename automatically
Miscellaneous
Keep broken files
Display files in Explorer
Save settings
Desktop
Downloads
Gallery
AKSHAYA - Personal
Desktop
Documents
Music
Pictures
Videos
Redmi Note 13 5G
Downloads
AKSHAYA D
This PC
Local Disk (C:)
Libraries
Network
Adaptive_Decision_Intelligence
akshaya.html
OK
Cancel
Help



RESULT:

Thus, a text file was appended to an image file using Command Prompt, producing C.jpeg. A ZIP archive containing a text file was also appended to an audio file using WinRAR and Command Prompt, producing newfile.mp3.

EXPERIMENT 10

EXTRACT EXIF DATA FROM IMAGE FILES USING EXIFREADER SOFTWARE

AIM:

To extract and examine Exchangeable Image File Format (EXIF) metadata from an image file using ExifReader software.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- Software: ExifReader
- An image file in JPEG or JPG format

THEORY:

EXIF is a standard used to store metadata associated with digital images. This information may be recorded by digital cameras, smartphones, and other imaging devices.

EXIF metadata may contain details such as:

- **CAMERA MAKE AND MODEL:** Information about the device used to capture the image.
- **DATE AND TIME:** The recorded date and time associated with the image, if available.
- **CAMERA SETTINGS:** Information such as shutter speed, aperture, and exposure.
- **FLASH INFORMATION:** Indicates whether the camera flash was used, if recorded.
- **IMAGE DIMENSIONS:** The width and height of the image.
- **GPS INFORMATION:** Geographic coordinates, if the device recorded them and the metadata remains available.

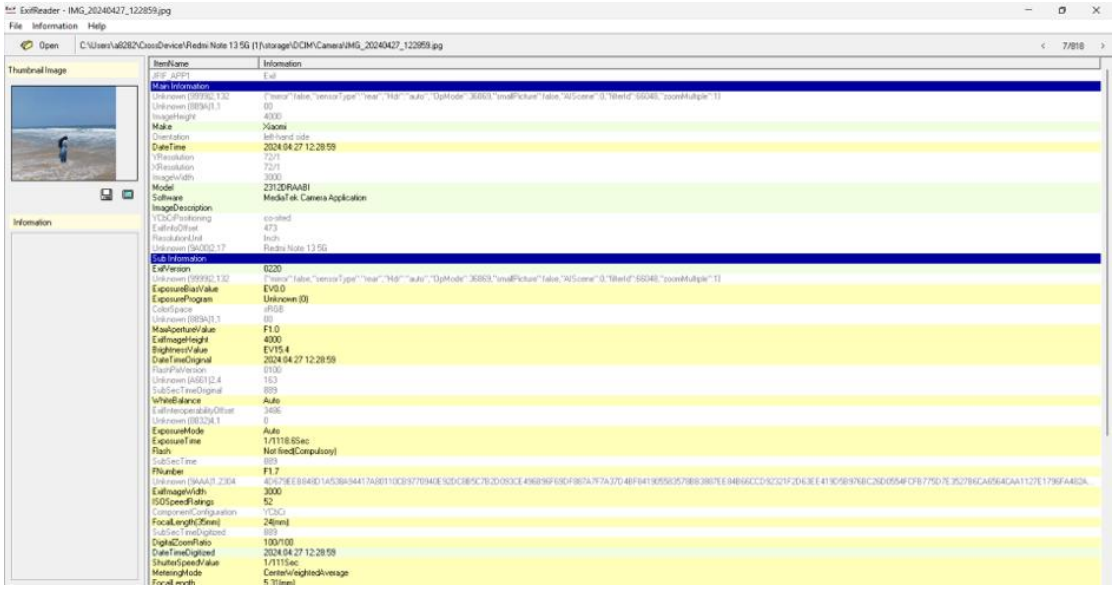
ExifReader is a Windows-based utility used to display EXIF metadata associated with supported image files. This information can help investigators examine images during a digital forensic investigation.

PROCEDURE:

1. Download and prepare ExifReader on the Windows computer.
2. Launch the ExifRead.exe application.
3. Click the **OPEN** button in the application window.
4. Browse to the location of the image file to be examined.
5. Select a JPEG or JPG image file and open it.
6. Observe the selected image displayed in the application.
7. Examine the EXIF metadata displayed by the software.
8. Identify the available details, such as camera information, image dimensions, date and time, and camera settings.
9. Record the relevant metadata for forensic examination.
10. Capture a screenshot showing the image and its available EXIF information.

OBSERVATION:

ExifReader displays the selected image and the EXIF metadata available within the file.



RESULT:

Thus, EXIF metadata was examined using ExifReader software to identify the available information associated with the selected image file.